

Lecture 4

Intensity Transformations and Spatial Filtering

空间的?

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Outline

What is image enhancement?

Why is it necessary? Why is it important?

Common methods for image enhancement in the spatial domain

- Point method

- Global method

- Spatial filtering method

What is image enhancement?

- ▶ A process that sharpens desired features in images
- ▶ What are desired features?
 - ▶ Highly subjective term
 - ▶ Varies from application to application
- ▶ How to evaluate the effect of enhancement?
 - ▶ Visual inspection is the major method
 - ▶ No real quantitative criterion available
 - ▶ Criterion changes as application tasks
 - ▶ Even for the same type of application, try and error approaches are often used to figure out the best method

Main Contents

What is image enhancement?

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Common methods for image enhancement in the spatial domain

Point method *don't care the position!*

Global method *histogram*

Spatial filtering method

What is Image Enhancement?

- ▶ Image enhancement very much related to feature extraction *? dilation?*
 - ▶ A preprocessing step for feature extraction, providing 'better' input for other automated image processing techniques
 - ▶ Improving the interpretability or perception of information in images for human viewers
 - ▶ Different transforms are common tools to look at images at different perspectives
- ▶ It will be nice to have a standardized, quantitative problem formulation and evaluation criterion for this topic. ✓

Common Enhancement Methods

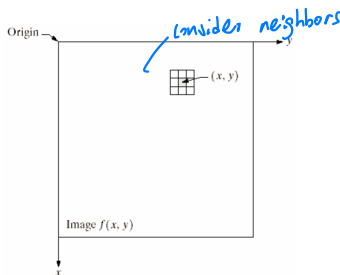
- ▶ What methods are available?
- ▶ How do these methods work?
- ▶ Applications
- ▶ Limitations and possible improvements

Enhancement Methods in the Spatial Domain

- ▶ What can we do in the spatial domain?
 - ▶ Image function: $f(x, y)$ — structured arrangement of function values
 - ▶ Methods work on intensity value f directly
 - ▶ Methods work on both (x, y) and f
- ▶ Classification of common spatial methods
 - ▶ Point operation
 - ▶ Spatial operation (Filtering and transforms)

Visualization and Mathematical Formulation of Spatial Methods

- General description of spatial methods: $g(x, y) = T[f(x, y)]$. *image in, image out*
- Operating range of T determines the classification of spatial method
- Visualization:



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Common Point Operation Methods

- ▶ Neighborhood of T : 1×1 pixel *Random operation*
 - ▶ For simplification: $s = T(r)$ or $v = T(u)$
 $\Rightarrow$ Everything happens on the intensity axis
 - ▶ Important: every pixel for himself spatial information completely lost!
- Assume that f is the clown image and T is a random function and apply $g = T(f)$:



Common Point Operation Methods

► Method 1: Clipping and Thresholding

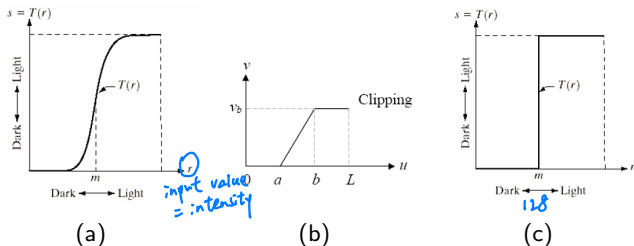
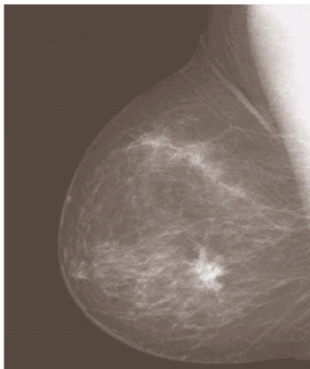


Figure: Clipping and Thresholding

Common Point Operation Methods: II

- ▶ **Method 2:** Image negatives. *flip black and white*
- ▶ Method detail: $s = L - 1 - r$ for $r \in [0, L - 1]$.
- ▶ enhance white or gray detail embedded in dark regions

super bright $\rightarrow$ super dark



Common Point Operation Methods: III

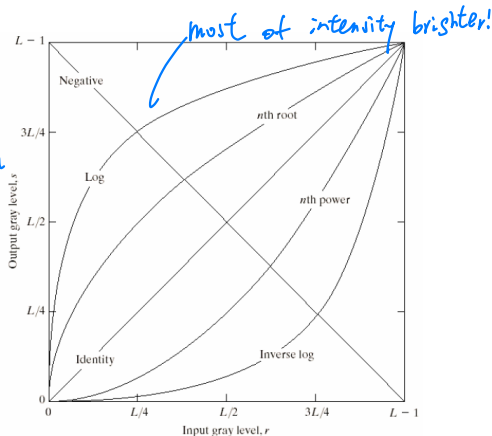
► Method 3:

Log Transformations

$$s = c \log(1 + r)$$

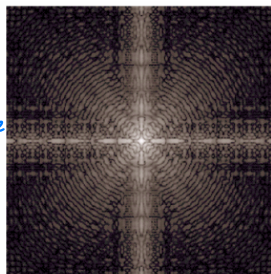
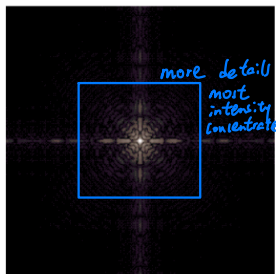
specific design function

expand the value of
dark pixels and
compress the bright
ones.



Example of Log Transformation

- Make dark pixels visible in the display:
Intensity value ranges from 0 to 1.5×10^6



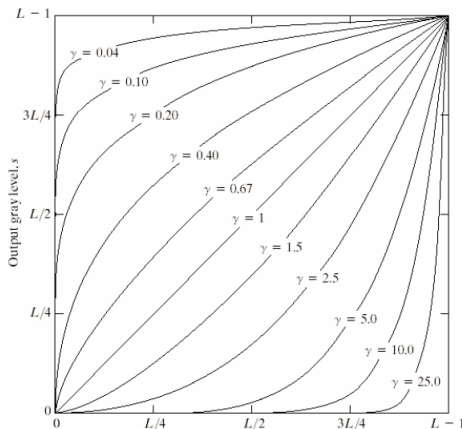
log transformation
 $c \log(1+I)$

Common Point Operation Methods: IV

► Method 4:

Power-Law Transforms:

$$s = cr \quad \text{or} \quad s = c(r + \epsilon)^{\gamma}$$



make sure that
 γ is consistent

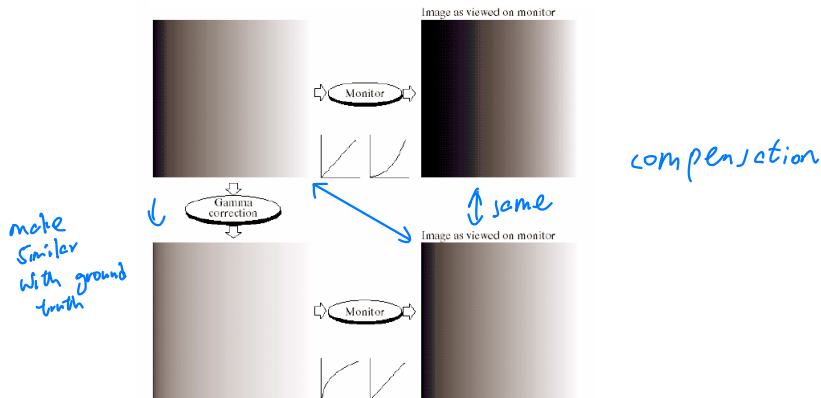
Why power laws are popular?

long time ago

- ▶ A cathode ray tube (CRT), for example, converts a video signal to light in a nonlinear way. The light intensity I is proportional to a power (γ) of the source voltage V_S
- ▶ For a computer CRT, γ is about 2.2
- ▶ Viewing images properly on monitors requires γ -correction

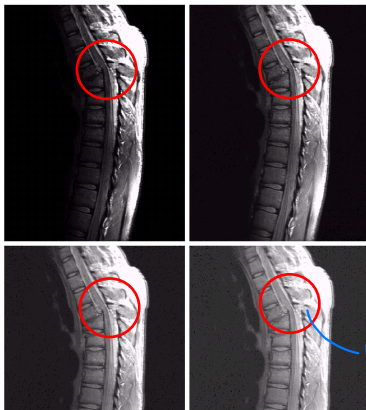
Example I of Power-Law Transformation

- ▶ The gamma correction of Cathode Ray Tube (CRT): intensity-to-voltage: a power function
- ▶ Display an image accurately on a computer screen



Example II of Power-Law Transformation

- ▶ Contrast Enhancement: $s = cr^\gamma$ with $c = 1$ and $\gamma = 0.6, 0.4, 0.3$
- ▶ More details become visible, but reduce contrast and tend to have a "washed-out" appearance.

a b
c d**FIGURE 3.8**

(a) Magnetic resonance (MR) image of a fractured human spine.

(b)–(d) Results of applying the transformation in Eq. (3.2-3) with $c = 1$ and $\gamma = 0.6, 0.4,$ and 0.3 , respectively.

(Original image for this example courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

more comparable!
that we can look!

Power-Law Transformation

- Contrast Enhancement: $c = 1$ and $\gamma = 3.0, 4.0, 5.0$

FIGURE 3.9

(a) Aerial image.
(b)–(d) Results of applying the transformation in Eq. (3.2-3) with $c = 1$ and $\gamma = 3.0, 4.0$, and 5.0 , respectively. (Original image for this example courtesy of NASA.)

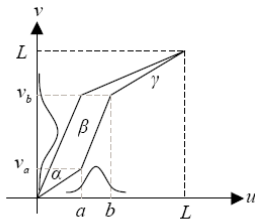


more darker
comparison ↑
much more clear :)!.

Piecewise-Linear Transformation: I

- Contrast stretching: with $\alpha = \frac{v_a}{a}, \beta = \frac{v_b - v_a}{b - a}, \gamma = \frac{L - v_b}{L - b}$

$$v = \begin{cases} \alpha u, & 0 \leq u < a, \\ \beta(u - a) + v_a, & a \leq u < b, \\ \gamma(u - a) + v_b, & b \leq u < L. \end{cases}$$



This region stretched.

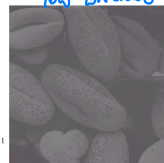
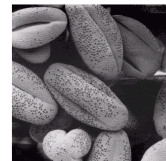
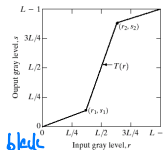
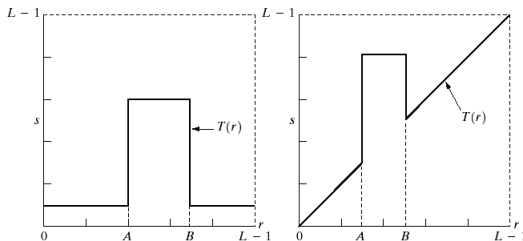


FIGURE 3.10
Contrast stretching.
(a) Form of transformation function. (b) A low-contrast image. (c) Result of contrast stretching. (d) Result of thresholding. (Original image courtesy of Dr. Roger Heady, Research School of Biological Sciences, Australian National University, Canberra, Australia.)

- expand the range of intensity

Piecewise-Linear Transformation: II

- Gray-level slicing: Highlight a specific range of interest



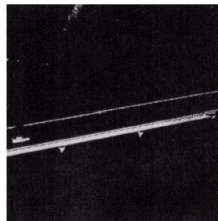
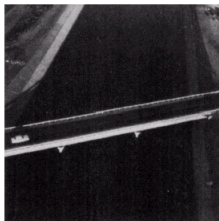
a	b
c	d

FIGURE 3.11

(a) This transformation highlights range $[A, B]$ of gray levels and reduces all others to a constant level.

(b) This transformation highlights range $[A, B]$ but preserves all other levels.

(c) An image.
(d) Result of using the transformation in (a).



Piecewise-Linear Transformation: III

- Bit-plane slicing: highlight contribution by specific bits

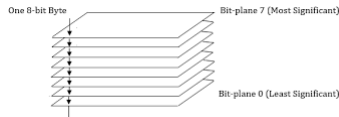
$$u = k_1 k_2 k_3 k_4 k_5 k_6 k_7 k_8$$

MSB

LSB

$$v = \begin{cases} L, & \text{if } n\text{-th significant bit} = 1 \\ 0, & \text{otherwise} \end{cases}$$

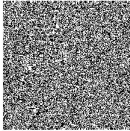
- The idea of basis image representation



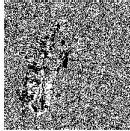
$$U = k_1 2^7 + k_2 2^6 + \dots + k_8$$

Example of Bit-plane Slicing

LSB Bit Plane



2nd Bit Plane



3rd Bit Plane



4th Bit Plane



5th Bit Plane



6th Bit Plane



7th Bit Plane



MSB Bit Plane



big picture'

Main Contents

What is image enhancement?

Why is it necessary? Why is it important?

Common methods for image enhancement in the spatial domain

Point method

Global method — care overall intensity distribution

Spatial filtering method

Histogram Based Method: Global Method

- ▶ A histogram is a graph that counts the number of occurrences of data points in a series of ranges or bins. *↳ normalize at last*
- ▶ What is a bin?
a bin = a container that holds all data points whose intensity values fall into the corresponding interval
- ▶ A normalized histogram is an estimate of the probability of occurrence of intensity

Example: Grade for 50 students

55	77	76	94	65
73	64	75	78	75
97	69	82	82	85
67	83	80	91	70
91	81	66	68	71
88	48	63	63	69
71	92	81	72	82
72	94	74	71	83
58	71	60	88	91

► Build a histogram

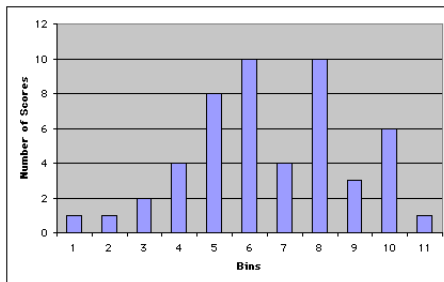
To make a histogram, we count up the number of data points between, 45 and 50, 50 and 55, 55 and 60, and so forth up to 95 to 100. Each of these ranges defines a 'bin' and the value in the 'Score' row reports the highest value in the bin:

Score	50	55	60	65	70	75	80	85	90	95	100
No. of Scores	1	1	2	4	8	10	4	10	3	6	1

[50, 54]

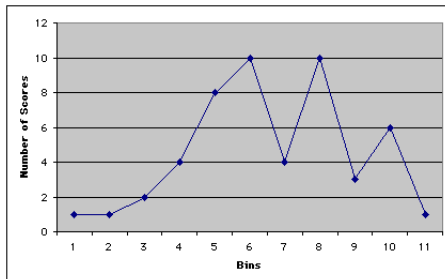
Plot a Histogram

- The graph of this table is the *histogram* where bin #1 corresponds to scores of 46 – 50 up to bin #11 which tells us that one student had a score between 96 and 100 points.



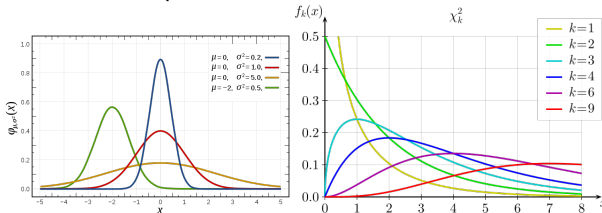
Alternative Plot of Histogram

- These rectangles are often omitted when the number of elevation ranges is large (since it makes the graph very messy).



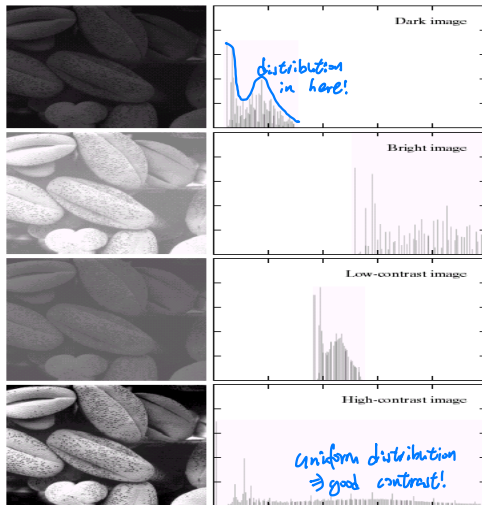
Information Conveyed by Histograms

- General shape of the frequency distribution
 - Normal, Chi-square, etc.



- Distribution is symmetric or skewed
- Modality - unimodal, bimodal, or multimodal

Images and Their Histograms



but not
always work!!!

Cannot do the opposite!

Critical Parameter in Histogram

- ▶ Bin width (the interval)
- ▶ Bin width too wide, what will happen?
- ▶ Bin width too small, what will happen? *lose the shape of pdf*
- ▶ How to choose the optimal bin width?
↳ silverman's rule

Histogram Method: Global Method

- ▶ Global method:

Not only the pixel itself, but the statistics in the **entire** image

- ▶ How do we use histogram information?

Focus on the distribution and the relative span range

- ▶ Different ways of modifying histogram

- ▶ Histogram equalization

- ▶ Histogram specification

↪ change find transfer function T such that output pdf is uniform
histogram we want, find T

Back to previous example

The following two images



have the same histograms

↳ cannot tell u
how image looks like

Histogram Equalization

- Idea: make histogram uniform distributed

Theorem *output image intensity*

Define $s = T(r)$ as the cumulative distribution function (CDF) of r , then the probability density function of s , $p_s(s)$, is always a uniform function.

- Any image can be equalized using this method

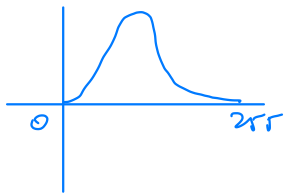
Histogram equalization:

Define

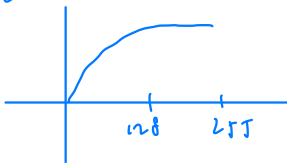
$$s_k = T(r_k) = \sum_{j=0}^k \frac{n_j}{n}$$

discrete version!
What in pixels are just discrete versions!

$p_d f(r)$



cdf



$p_r(r)$ pdf of r

$p_s(s)$ pdf of s

find $T \Rightarrow s = T(r)$

$$s = T(r) = (L-1) \int_0^r \underbrace{p_r(w) dw}_{\text{CDF of } w}$$

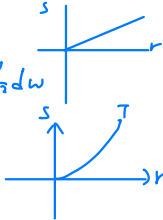
$$\begin{aligned} \frac{ds}{dr} &= \frac{dT(r)}{dr} = (L-1) \frac{d}{dr} \int_0^r p_r(w) dw \\ &= (L-1) \underbrace{p_r(r)}_{\text{intensity value of } r!} \end{aligned}$$

probability manipulation

$$\begin{aligned} P_s(s) &= P_r(r) \left| \frac{dr}{ds} \right| \\ &\quad \uparrow \text{inverse function} \\ &= P_r(r) \frac{1}{(L-1) P_r(r)} \\ &= \frac{1}{L-1} \text{ fixed number} \end{aligned}$$

$$p_r(r) = \frac{2r}{(L-1)^2}$$

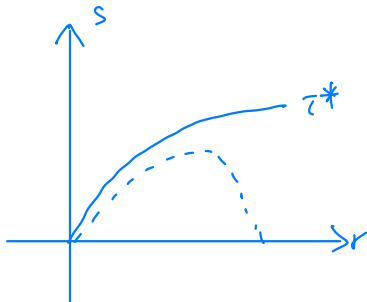
$$s = T(r) = (L-1) \int_0^r \frac{2w}{(L-1)^2} dw$$



Validation

$$p(s) = p(r) \frac{dr}{ds}$$

$$= \frac{2r}{(L-1)^2} \cdot \frac{1}{\frac{ds}{dr}} = \frac{2r}{(L-1)^2} \cdot \frac{1}{\frac{(2r)}{L-1}} = \frac{1}{L-1}$$



Accumulate the number



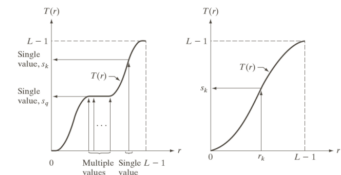
$$p(n_k) = \frac{n_k}{M/N}$$

(make use within $z_0, 1$)

$$S_k = \underbrace{(b-1)}_{255} \sum_{i=0}^k p(n_i)$$

Histogram Equalization

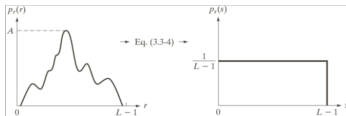
- ▶ apply a monotone transform resulting in an approximately uniform histogram



a b

FIGURE 3.17

(a) Monotonically increasing function, showing how multiple values can map to a single value. (b) Strictly monotonically increasing function. This is a one-to-one mapping, both ways.

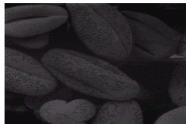


a b

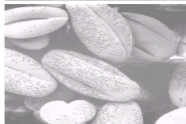
FIGURE 3.18 (a) An arbitrary PDF. (b) Result of applying the transformation in Eq. (3.3-4) to all intensity levels, r . The resulting intensities, s , have a uniform PDF, independently of the form of the PDF of the r 's.

After Histogram Equalization

(1)



(2)



(3)

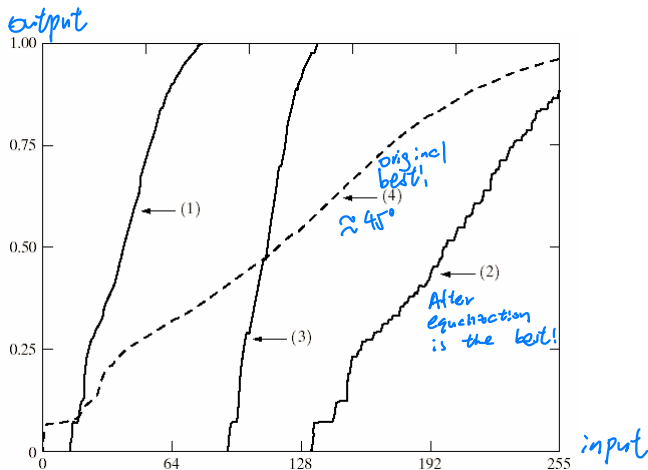


(4)



relatively
flatter
uniform
distribution!

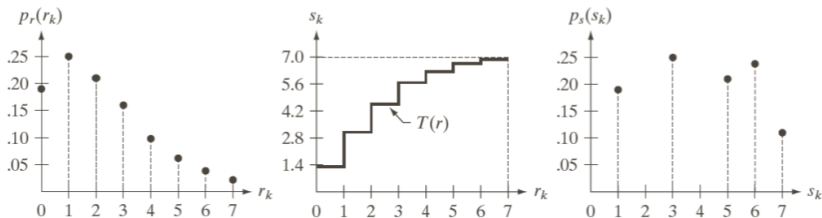
Transformation Functions: Corresponding to Previous Images



How and why does it work

3 bits

resolution poor!



a b c

FIGURE 3.19 Illustration of histogram equalization of a 3-bit (8 intensity levels) image. (a) Original histogram. (b) Transformation function. (c) Equalized histogram.

Details on the white board.

Y	$P(Y)$		X	$S = \text{round off (7pts)}$	$P(S)$
0	0.19	0.19	1.33	1	0.19
1	0.25	0.44	3.08	3	0.25
2	0.21	0.65	4.55	5	0.21
3	0.16	0.81	5.67	6	0.24
4	0.08	0.89	6.23	6	
5	0.06	0.95	6.65	7	0.09
6	0.03	0.98	6.86	7	
7	0.02	1	7	7	0.02

Approximation: ✓

~~✓~~
X

X

✓

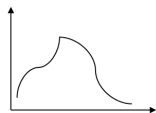
because no
ground truth,
need approximate

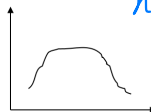
not uniform
distribution?

But better!

Histogram Specification

- more general case !
 an image's histogram is transformed according to a desired function
- Transforming the intensity values so that the histogram of the output image approximately matches a specified histogram

histogram₁

$$S^{-1} * T$$
histogram₂

name output distribution

find T?

highlight part

T₁



T



S



T_2

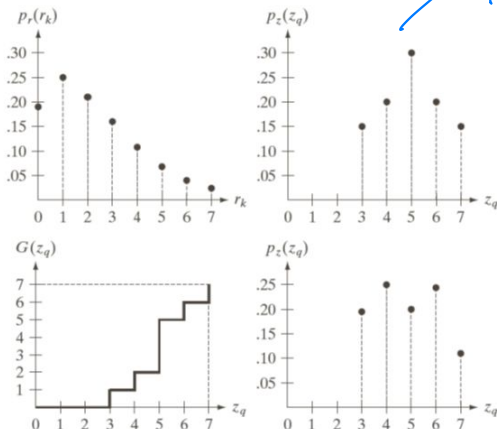


Histogram Specification

- ▶ How does it work?
 - ▶ Do the histogram equalization for the input image and obtain an input-output intensity transform function f_1 .
 - ▶ Do the histogram equalization for the output image (with known histogram) and obtain the second input-output intensity transform function f_2 .
 - ▶ Take the inverse of the second input-output intensity transform function $g_2 = \text{inv}(f_2)$.
 - ▶ Cascade g_2 and $f_1 : h = g_2(f_1)$
 - ▶ Transform the input image based on h to obtain the output image with the specified histogram.

Histogram specification: an example

Details on the white board.



a b
c d

FIGURE 3.22

(a) Histogram of a 3-bit image. (b) Specified histogram. (c) Transformation function obtained from the specified histogram. (d) Result of performing histogram specification. Compare (b) and (d).

output histogram may not be same as the desired one!

S cdf $\times 7$	$\text{cdf}(z)$	$P(z)$	z
$0 = 0$	0	0	0
$0 = 0$	0	0	1
$0 = 0$	0	0	2
$1.05 = 1$	0.15	0.15	3
$2.45 = 2$	0.35	0.2	4
$4.55 = 5$	0.65	0.3	5
$5.55 = 6$	0.85	0.2	6
7	1	0.15	7

$\uparrow$ S^* $\uparrow$
 $\uparrow$ $(S^{-1})^*$ $\uparrow$

- if cannot find number
choose closest to map'.

$$p(r) = \frac{2r}{(L-1)^2} \Rightarrow p(z) = \frac{2z^2}{(L-1)^3}$$

$$\begin{aligned} S &= \tau(r) \\ &= (L-1) \int_0^r \frac{2w}{(L-1)} dw \quad G(z) = \tau(z) \\ &= (L-1) \int_0^z \frac{2w^2}{(L-1)^3} dw \\ &= \frac{z^3}{(L-1)^2} = S \end{aligned}$$

$z = [(L-1)^2 S]^{\frac{1}{3}}$
 $z = [(L-1)^2 r^2]^{\frac{1}{3}}$

$$z = T^{-1}[\tau(t)]$$

T^{-1} is inverse function

A Few Comments on Histogram

- ▶ A global method: work on the entire image
⇒ local histogram to increase localization ability
- ▶ May not specific enough: *not good enough!*
 - ▶ Just an intensity distribution
 - ▶ Neighborhood information largely weakened
don't know what image looks like!
- ▶ Use of Histogram statistics in enhancement: e.g. n -th moment information

mathematical/logical operation on images

- ▶ Addition: averaging for noise removal
- ▶ Subtraction:
 - ▶ remove background
 - ▶ image matching
 - ▶ moving object tracking
- ▶ Multiplication:
 - ▶ superimposing of texture on an image
 - ▶ convolution and correlation
- ▶ And or Or: remove unnecessary area of an image

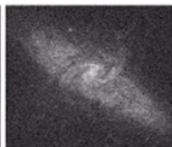
Image averaging

- ▶ The noise on the image can be removed out by taking more noisy images of the same object and average out

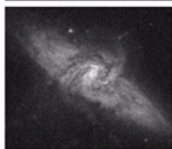
Original image



Noisy image



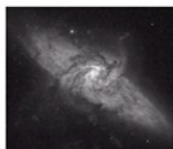
Average white noise!



Result of
averaging using
8 noise samples



Using 16 noise
samples



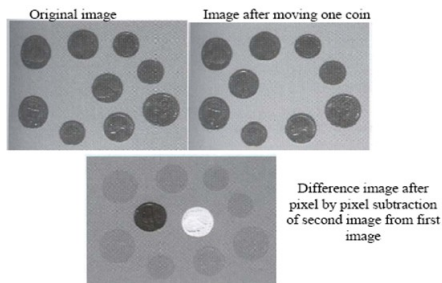
Using 64 noise
samples



Using 128 noise
samples

Image Subtraction

- ▶ Motivation: remove background/uninteresting part
- ▶ Math form: $g(x, y) = f(x, y) - h(x, y)$
- ▶ Example:



Multiplication

- image multiplication $Q(i, j) = P_1(i, j) \times P_2(i, j)$

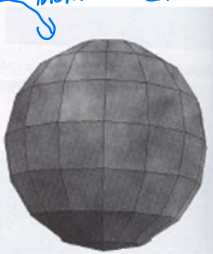
Multiplication of images can be used for superimposing texture on an image



Smooth spherical surface
image



Texture to be
superimposed



output image

Main Contents

What is image enhancement?

Why is it necessary? Why is it important?

Common methods for image enhancement in the spatial domain

Point method

Global method

Spatial filtering method

} ← in between

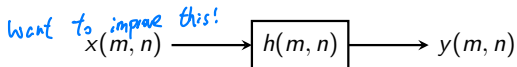
Spatial Filtering

- Spatial correlation

$$y(m, n) = \sum_{s=-a}^a \sum_{t=-b}^b h(s, t) x(m + s, n + t)$$

Spatial convolution

$$y(m, n) = \sum_{s=-a}^a \sum_{t=-b}^b h(s, t) x(m - s, n - t)$$



- Key element: design of the mask
- Root: in the frequency domain
- Computation type: **Inner product**. WHY?

Smooth Filtering

Consider the output pixel is positioned at the center

$$\frac{1}{9} \times$$

1	1	1
1	1	1
1	1	1

Box filter all coefficients are equal

Consider mask size:

$$m \times n$$

$$w_i = \frac{1}{mn}, i = 1, \dots, mn$$

$$\frac{1}{16} \times$$

1	2	1
2	4	2
1	2	1

Weighted average give more (less) weight to pixels near (away from) the output location

Smoothing Masks in the Spatial Domain

- Average mask:

$$\begin{array}{c}
 \xrightarrow{\quad} n \\
 \begin{array}{cc}
 0 & 1 \\
 \hline
 0 & \begin{array}{|c|c|} \hline \frac{1}{4} & \frac{1}{4} \\ \hline \end{array} \\
 \downarrow m \\
 1 & \begin{array}{|c|c|} \hline \frac{1}{4} & \frac{1}{4} \\ \hline \end{array}
 \end{array}
 \Rightarrow y(m,n) = \frac{1}{4} [x(m-1,n) + x(m,n) + x(m,n-1) + x(m-1,n-1)]
 \end{array}$$

Averaging

- Useful to reduce noise. Why?
- How does it look like in the spatial domain?

Example

Average out the noise!

Let $\tilde{x}(m, n) = x(m, n) + \eta(m, n)$, then

$$\begin{aligned} y(m, n) &= \tilde{x}(m, n) * h(m, n) \\ &= x(m, n) * h(m, n) + h(m, n) * \eta(m, n). \end{aligned}$$

Assume that $\eta(m, n) \sim N(0, \sigma^2)$, then the variance of modified noise $\hat{\eta}(m, n) = h(m, n) * \eta(m, n)$ is usually smaller, e.g.

$$\begin{aligned} \hat{\eta}(m, n) &= \frac{1}{4} [\eta(m, n) + \eta(m-1, n) + \eta(m, n-1) + \eta(m-1, n-1)] \\ \Rightarrow \text{variance of } \hat{\eta}(m, n) &= \frac{\sigma^2}{4}. \end{aligned}$$

Smooth Filtering: example

tradeoff between reduce noise and preserve input details!

Original image
size: 500 x 500



Smoothed by
3 x 3 box filter



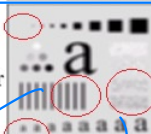
Smoothed by
5 x 5 box filter



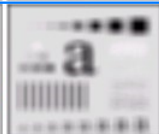
Smoothed by
9 x 9 box filter



Smoothed by
15 x 15 box filter



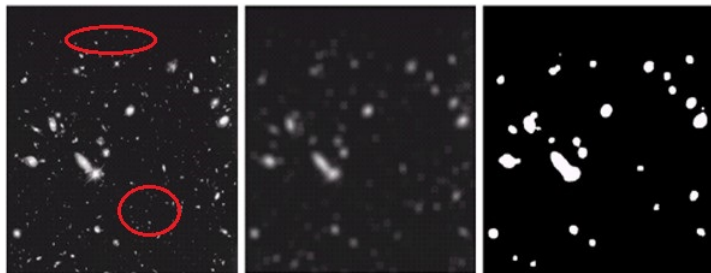
Smoothed by
35 x 35 box filter



Also blur!

noisy pattern gone!

Smooth Filtering: example



a b c

FIGURE 3.36 (a) Image from the Hubble Space Telescope. (b) Image processed by a 15×15 averaging mask. (c) Result of thresholding (b). (Original image courtesy of NASA.)

Order-statistics filtering

Nonlinear filter, based on order of the gray levels in the mask:

Median, Max and Min filtering

- ▶ Median mask: replace mean with median Remarks:
 - ▶ Median filter is nonlinear

$$\text{median}\{x(m) + y(m)\} \neq \text{median}\{x(m)\} + \text{median}\{y(m)\}$$

- ▶ Useful for reducing shot noise and binary noise, but not useful for Gaussian noise.
- ▶ When number of noisy pixel $\geq$ total number of pixels in the mask, performance is poor.

Example of Median Filtering

White
Salt and pepper noise



Depends on my
experience!



Always not extreme values!

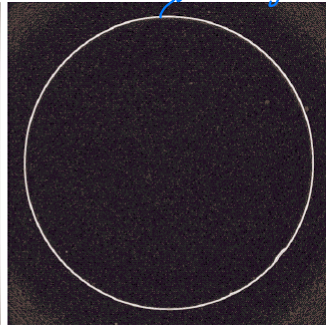
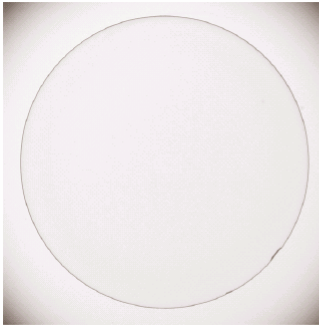
Edge Enhancement Masks

► Gradient:

Grey change $\Rightarrow$ extract high frequency signals!

-1	0	0	-1
0	1	1	0

detect edge!!!



More Edge Enhancement Masks

	0°	45°																		
Sobel	<table border="1"> <tr><td>-1</td><td>0</td><td>1</td></tr> <tr><td>-2</td><td>0</td><td>2</td></tr> <tr><td>-1</td><td>0</td><td>1</td></tr> </table>	-1	0	1	-2	0	2	-1	0	1	<table border="1"> <tr><td>0</td><td>1</td><td>2</td></tr> <tr><td>-1</td><td>0</td><td>1</td></tr> <tr><td>-2</td><td>-1</td><td>0</td></tr> </table>	0	1	2	-1	0	1	-2	-1	0
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-2	0	2																		
-1	0	1																		
0	1	2																		
-1	0	1																		
-2	-1	0																		
Kirsch	<table border="1"> <tr><td>-3</td><td>-3</td><td>5</td></tr> <tr><td>-3</td><td>0</td><td>5</td></tr> <tr><td>-3</td><td>-3</td><td>5</td></tr> </table>	-3	-3	5	-3	0	5	-3	-3	5	<table border="1"> <tr><td>-3</td><td>5</td><td>5</td></tr> <tr><td>-3</td><td>0</td><td>5</td></tr> <tr><td>-3</td><td>-3</td><td>-3</td></tr> </table>	-3	5	5	-3	0	5	-3	-3	-3
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Robinson	<table border="1"> <tr><td>-1</td><td>0</td><td>1</td></tr> <tr><td>-1</td><td>0</td><td>1</td></tr> <tr><td>-1</td><td>0</td><td>1</td></tr> </table>	-1	0	1	-1	0	1	-1	0	1	<table border="1"> <tr><td>0</td><td>1</td><td>1</td></tr> <tr><td>-1</td><td>0</td><td>1</td></tr> <tr><td>-1</td><td>-1</td><td>0</td></tr> </table>	0	1	1	-1	0	1	-1	-1	0
-1	0	1																		
-1	0	1																		
-1	0	1																		
0	1	1																		
-1	0	1																		
-1	-1	0																		

First and Second order derivatives

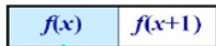
how brain
detects
detail?

some 神经网络
detects the
specific

how to think the values?
About brightness changes?

next neighbour

$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$



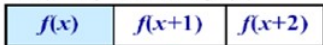
Position for the
output pixel

2 neighbors

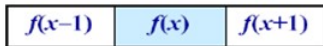
$$\frac{\partial^2 f}{\partial x^2} = f'(x+1) - f'(x)$$

$$= [f(x+2) - f(x+1)] - [f(x+1) - f(x)]$$

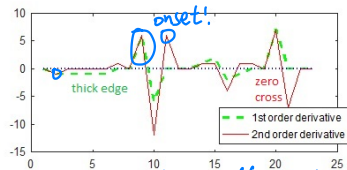
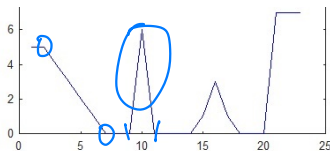
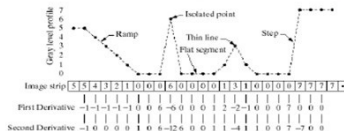
$$= f(x+2) - 2f(x+1) + f(x)$$



$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$



Example for discrete derivatives



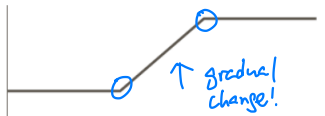
1st derivative:

- ▶ zeros in areas of constant intensity
- ▶ nonzeros at the onset of an intensity step or ramp
→ graduate change!
- ▶ nonzero along intensity ramps

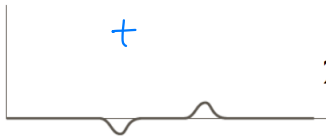
2nd derivative:

- ▶ zeros in areas of constant intensity
- ▶ nonzeros at the onset AND END of an intensity step or ramp
- ▶ ZEROS along intensity ramps

How do we utilize 2nd order derivative



Original signal



2nd derivative

better for change values!
what start what end



Original signal - 2nd derivative
enhance the edge!

Laplacian for Edge Enhancement

not just one direction but also another direction!

Laplacian operator

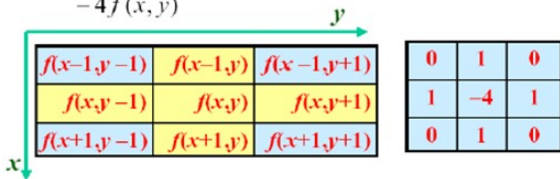
$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

Adding 4 neighbours!

$$\nabla^2 f = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$



Laplacian for Image Enhancement

0	1	0	1	1	1
1	-4	1	1	-8	1
0	1	0	1	1	1

0	-1	0	-1	-1	-1
-1	4	-1	-1	8	-1
0	-1	0	-1	-1	-1

a b
c d

FIGURE 3.39

(a) Filter mask used to implement the digital Laplacian, as defined in Eq. (3.7-4). (b) Mask used to implement an extension of this equation that includes the diagonal neighbors. (c) and (d) Two other implementations of the Laplacian.

*detect the different direction
treat all directions equally
important!*

Laplacian for Image Enhancement

≈ high pass filter!

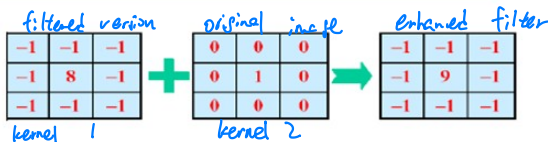
To obtain the enhanced image $\hat{p}$

$$g(x, y) = \begin{cases} f(x, y) - \nabla^2 f(x, y), & w_s < 0 \\ f(x, y) + \nabla^2 f(x, y), & w_s > 0 \end{cases}$$

w'_1	w'_2	w'_3
w'_4	w'_5	w'_6
w'_7	w'_8	w'_9

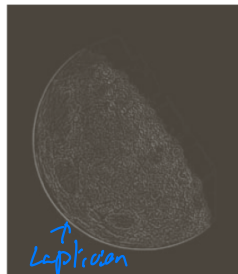
emphasise!

In this way, background tonality can be perfectly preserved while details are enhanced



⇒ do convolution then add!

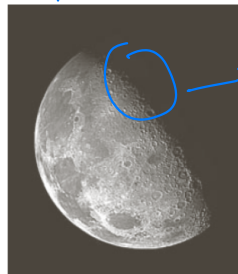
Laplacian for Image Enhancement: example



detect edges
 → only edges
 can be white!
 !!
 high pass filter

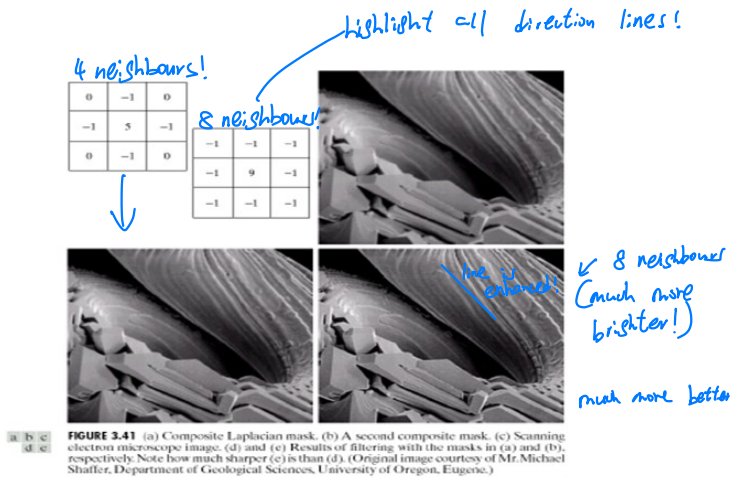
↑
 Laplacian

4 neighbours



≤ 8 neighbours
 → clear!

Laplacian for Image Enhancement: example



Summary on Spatial Masks

- ▶ The masks mimic the structure of interests in images → Why?
 - ▶ Remember convolution/inner product?
 - ▶ Filter response = similarity measure
- ▶ Very intuitive, but may be difficult to design or compare
- ▶ View in the frequency domain may be helpful
→ Topics in the next week

Reading Assignment

- ▶ Read Chapter 3 of the textbook